

Habiba Nassar



GitHub



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LinkedIn



Portfolio

Education

BSc Computer Science, The University of Edinburgh

Sep. 2024 – May. 2028

Predicted First Class Honours

Edinburgh, Scotland

Relevant Modules: Algorithms & Data Structures, Software Engineering and Professional Practice, Reasoning and Agents, Computer Systems

Technical Skills

Languages: Java, Python, C, SQL, JavaScript, HTML/CSS, MIPS Assembly, x86 Assembly

Frameworks & Libraries: React, JUnit, Mockito, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

Tools & Technologies: Git, GitHub, Docker, Kubernetes, TeamCity, Jira, Maven

Experience

Software Developer Intern

Summer 2026

SGX FX

Edinburgh, Scotland

- Optimised cache lookup performance by replacing a list with a **HashMap**, reducing lookup time from **40–60 seconds to around 100 ms**; the improvement was deployed to production and recognised during a team demo.
- Applied **Test-Driven Development (TDD)** by writing unit and integration tests for code changes, helping ensure the reliability of new and modified functionality.
- Collaborated in an Agile development team through sprint planning, daily stand-ups and code reviews, using GitHub, Jira, TeamCity, Docker and Kubernetes throughout the development workflow.

Teaching Support Provider (Tutor & Marker)

Sep. 2025 – Dec. 2025

University of Edinburgh School of Informatics

Edinburgh, Scotland

- Tutored Computer Science students in functional programming and computational logic through weekly tutorial sessions
- Marked coursework and provided constructive feedback to support students' understanding and academic progress
- Facilitated interactive tutorials, encouraging problem solving and active participation

IT Equipment Refurbishment Volunteer

Jul. 2025 – Oct. 2025

Turing Trust

Edinburgh, Scotland

- Refurbished donated desktop and laptop computers for deployment to schools in disadvantaged communities
- Diagnosed hardware faults, installed OSs, securely wiped storage devices, and tested systems for reuse
- Upgraded and replaced hardware components, including RAM and storage devices, while collaborating with volunteers to refurbish IT equipment efficiently

Projects

Kaprekar's Constant | GitHub

React, JavaScript, HTML, CSS

- Implemented Kaprekar's routine algorithm to repeatedly rearrange and subtract digits until reaching the constant, **6174**.
- Built input validation and error handling for 4-digit numbers, including handling leading zeroes throughout the calculation.
- Created a dynamic React interface that visualises each step of the routine based on user input.

Events Booking Application

Java, Maven, JUnit, Mockito, Git

- Developed a Java events booking application using object-oriented principles and the MVC design pattern, supporting user registration, event search, bookings, cancellations, and multiple user roles
- Developed unit and integration tests using JUnit and Mockito, covering edge cases and error handling
- Collaborated in an Agile team using Git/GitHub and code reviews, contributing across requirements elicitation, design, implementation, testing, and integration

Solar System Simulation | GitHub

Python, NumPy, Matplotlib

- Developed an object-oriented simulation in Python using NumPy and Beeman's method for planetary motion
- Used Matplotlib to analyse energy conservation and compare Beeman, Euler-Cromer, and Direct Euler methods
- Calculated planetary orbital periods and detected planetary alignments from simulated trajectories

Cache Simulator

C, MIPS Assembly

- Developed a configurable cache simulator in C supporting direct-mapped, set-associative, and two-level cache architectures
- Implemented cache size, block size and associativity with LRU replacement and write-back/write-allocate policies
- Used dynamic memory allocation and bit manipulation to simulate memory accesses and track cache hits and misses

Awards and Achievements

UG1 Programme Representative • HumanUtopia Hero • DofE Award • Higher Leadership Award